# 4.9 Holditch's Theorem

**WORKSHEET**

**The question**

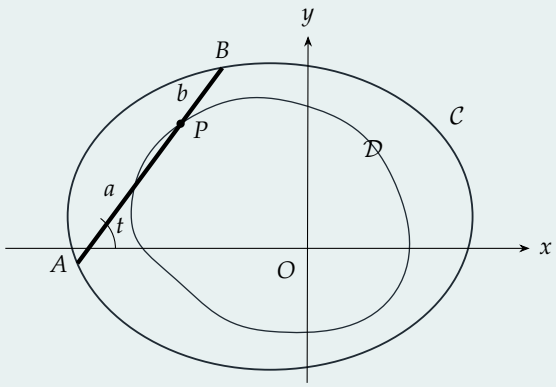
A movable point $P$ has cartesian coordinates $(x, y)$, where $x$ and $y$ are functions of $t$. The polar coordinates of $P$ with respect to the origin $O$ are $r$ and $\theta$. Starting with the expression

$$\tfrac{1}{2}\int r^2 \,\mathrm{d}\theta$$

for the area swept out by $OP$, obtain the equivalent expression

$$\tfrac{1}{2}\int \left(x\frac{\mathrm{d}y}{\mathrm{d}t} - y\frac{\mathrm{d}x}{\mathrm{d}t}\right)\mathrm{d}t. \qquad (*)$$

The ends of a thin straight rod $AB$ lie on a closed convex curve $\mathcal{C}$. The point $P$ on the rod is a fixed distance $a$ from $A$ and a fixed distance $b$ from $B$. The angle between $AB$ and the positive $x$ direction is $t$. As $A$ and $B$ move anticlockwise round $\mathcal{C}$, the angle $t$ increases from 0 to $2\pi$ and $P$ traces a closed convex curve $\mathcal{D}$ inside $\mathcal{C}$, with the origin $O$ lying inside $\mathcal{D}$, as shown in the diagram.




Let $(x, y)$ be the coordinates of $P$. Write down the coordinates of $A$ and $B$ in terms of $a$, $b$, $x$, $y$ and $t$.

The areas swept out by $OA$, $OB$ and $OP$ are denoted by $[A]$, $[B]$ and $[P]$, respectively. Show, using $(*)$, that

$$[A] = [P] + \pi a^2 - af$$

where

$$f = \tfrac{1}{2}\int_0^{2\pi} \left( \left( x + \frac{\mathrm{d}y}{\mathrm{d}t} \right) \cos t + \left( y - \frac{\mathrm{d}x}{\mathrm{d}t} \right) \sin t \right) \mathrm{d}t.$$

Obtain a corresponding expression for $[B]$ involving $b$. Hence show that the area between the curves $\mathcal{C}$ and $\mathcal{D}$ is $\pi ab$.

## Hints

Try the question on your own first. If you get stuck, read the hints *one at a time*, and go back to the question after each one. Hints marked *Pause and try* are the key steps.

**Ask yourself** Check the theorem in the simplest case. If $\mathcal{C}$ is a circle of radius $R$, what curve does $P$ trace, and what is the area of the ring?

### STAGE 1 Area from coordinates

**Ask yourself** With $x = r\cos\theta$ and $y = r\sin\theta$, all functions of $t$, compute $x\dot{y} - y\dot{x}$, where dots denote $\frac{\mathrm{d}}{\mathrm{d}t}$.

### STAGE 3 The area swept by $OA$

**Ask yourself** Apply $(*)$ to $A$. Expand $x_A\dot{y}_A - y_A\dot{x}_A$ and sort the terms by powers of $a$.

**Pause and try** Now find $[B]$. (*Hint:* $B$ is obtained from the formula for $A$ by replacing $a$ with what?)

### STAGE 4 Eliminating $f$

**Ask yourself** Both $A$ and $B$ go once round the same curve $\mathcal{C}$. What does that tell you about $[A]$ and $[B]$?

## Try this next

1. **The area of an ellipse.** Use $(*)$ with $x = p\cos t$, $y = q\sin t$ to show that an ellipse with semi-axes $p$ and $q$ has area $\pi pq$.

2. **The circle case.** Prove the intersecting chords theorem: if two chords of a circle meet at $P$, the products of the lengths of their two segments are equal. Deduce Holditch's theorem for a circle.

3. **The midpoint.** Show that $f = \pi(a - b)$ means that when $P$ is the midpoint of the rod, $f = 0$. What is the ring's area then, in terms of the rod's length $L$?

4. **A triangle.** HARDERUse $(*)$ to find the area of the triangle with vertices $(0,0)$, $(x_1, y_1)$, $(x_2, y_2)$, traversing its boundary, and show it equals $\frac{1}{2}|x_1y_2 - x_2y_1|$. This is the start of the "shoelace formula" for the area of any polygon.

## Further reading

- **Mark Levi, *The Mathematical Mechanic*** (Princeton University Press). Uses physical reasoning, with rods, wheels and springs, to prove mathematical results, in the spirit of this section.
- **H. M. Schey, *Div, Grad, Curl, and All That*** (Norton). An informal introduction to vector calculus, including Green's theorem.
- **Hamnet Holditch, "Geometrical theorem"**, *Quarterly Journal of Pure and Applied Mathematics* 2 (1858). The original note, a single short paragraph with none of the fine print.

The full guided solution, and what the question leads to at university, is in Section 4.9:
https://keithmcnulty.github.io/step_book_mcnulty/ch04/holditch.html